

Temperature, Depth and Salinity Measurements at Dana Point

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Problem Statement

The goal of our project was to design an autonomous, underwater vehicle (AUV) that is capable of taking temperature, depth and salinity measurements with reasonable accuracy. South Orange County Wastewater Authority has a wastewater treatment plant adjacent to Dana Point Harbor which can lead to untreated wastewater entering the recreational source of water if factors like precipitation cause overflow in sewage lines. Our AUV and its measurements can help with understanding the freshness of the water at Dana Point and serve as knowledge for advising safe recreation.

Impact

- Differences in temperature and salinity impact water density, which can create dead zones that can destroy ecosystems.
- Salinity measurements can also detect freshwater runoff and indicate areas that might be negatively influenced by human activity.
- Warmer temperature measurements can be used as an indicator of climate change.

Future Work

- Identifying a salinity sensor that can accurately measure salinity while minimizing or preventing corrosion during long-term use
- Minimizing error of the sonar sensor's measured depth by improving calibration, and testing in more realistic water conditions.
- Develop the sonar sensor to work at larger depths and over uneven surfaces, such as the ocean floor.

Autonomous Underwater Vehicle (AUV) Features

Electrical Schematic & Sensors

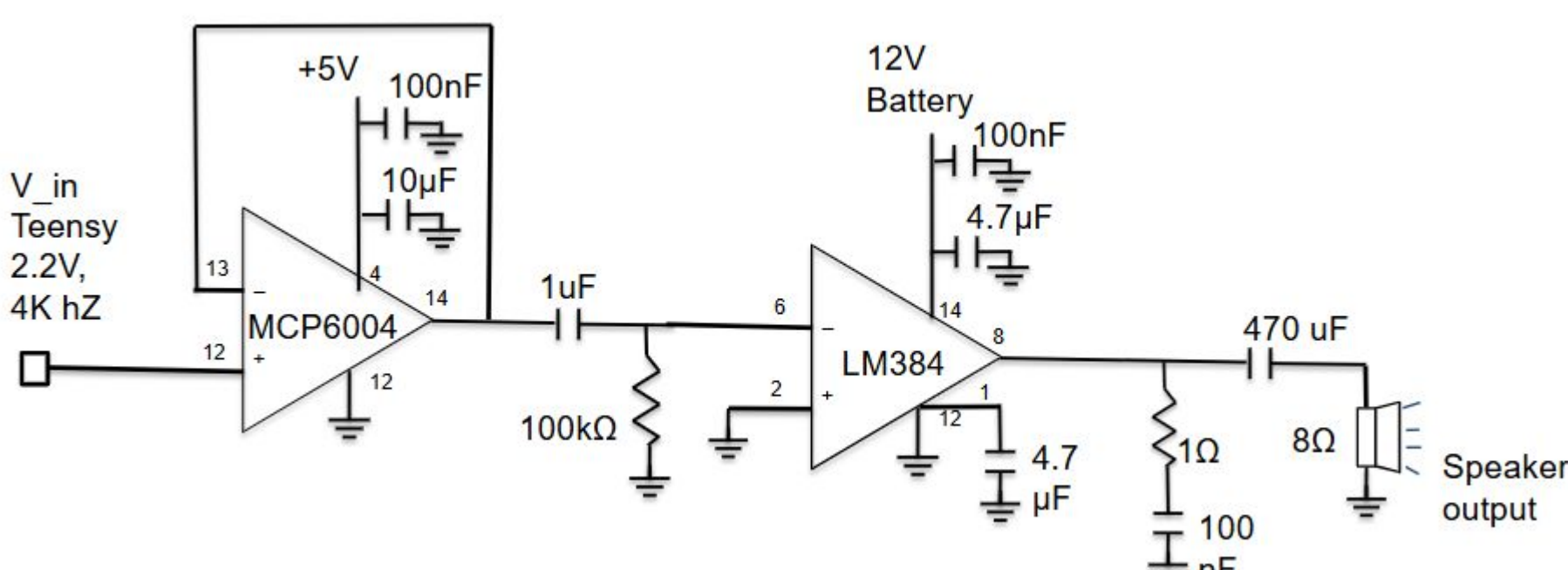


Figure 1. Speaker Schematic showing speaker that takes 2.2 Vpp 4 kHz pulse for 1ms and amplifies using LM384 amplifier

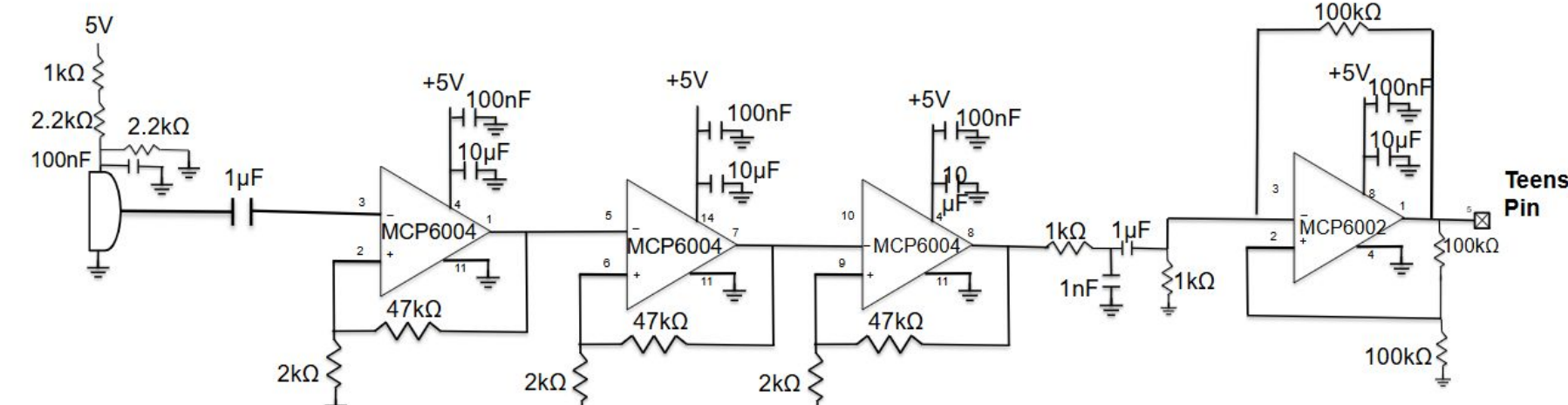


Figure 2. Microphone Schematic: Takes microphone input, sends to 3, 24x gain stages and filters out frequencies greater than 6.8kHz

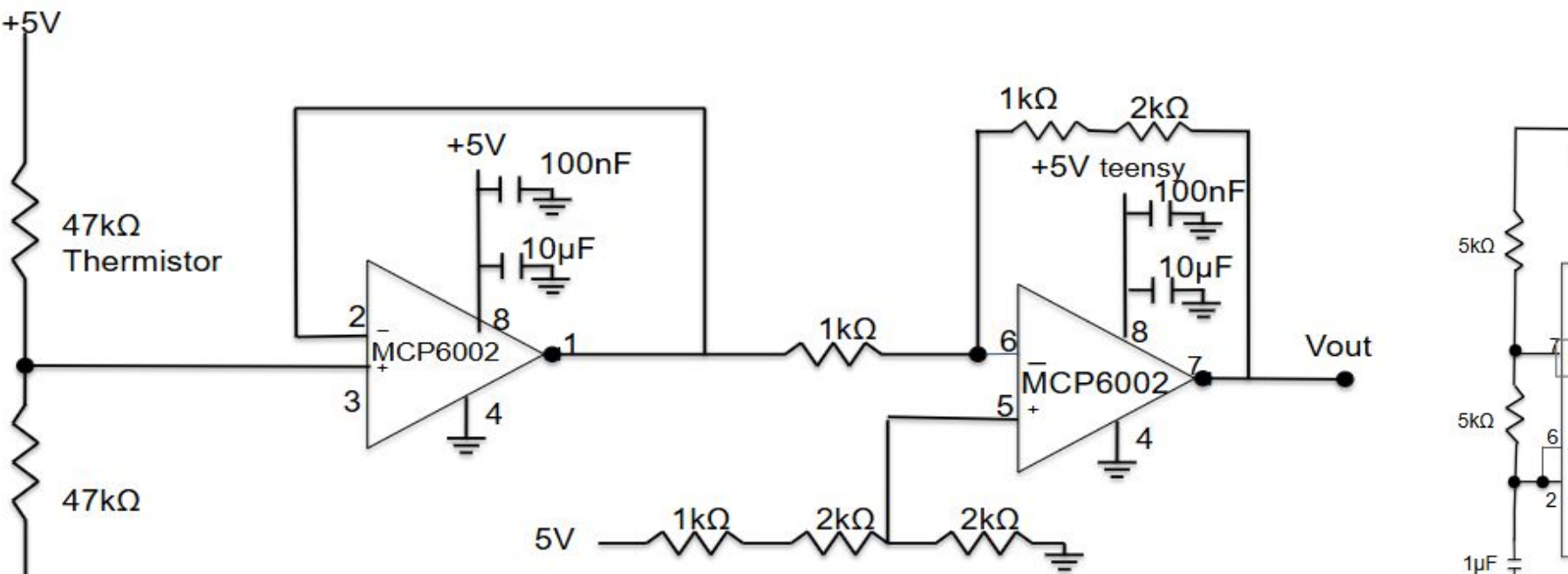


Figure 3. Thermistor Schematic: Voltage divider output shifted and scaled by inverting offset op-amp to 0.5-3.08V for temperatures 10-25 °C.

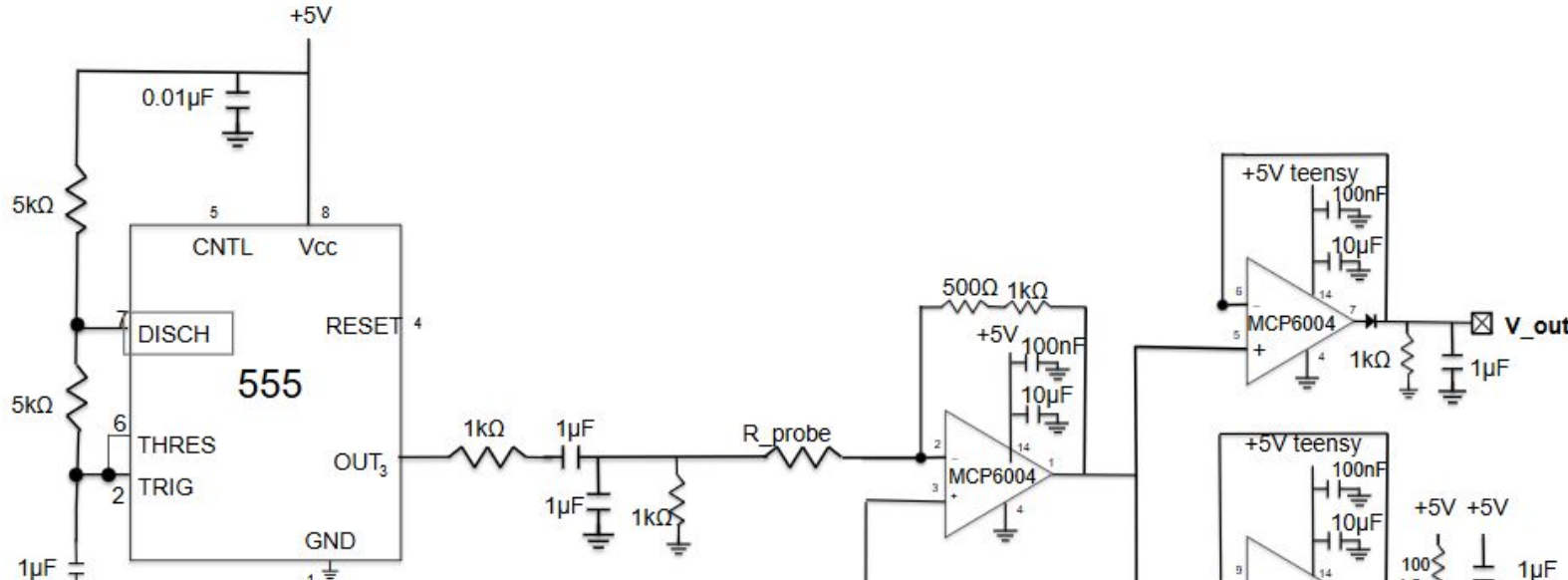


Figure 4. Salinity Schematic: 555 sends 200 Hz sine wave to probe, max and min detectors measure voltage difference

Mechanical Design & Integration

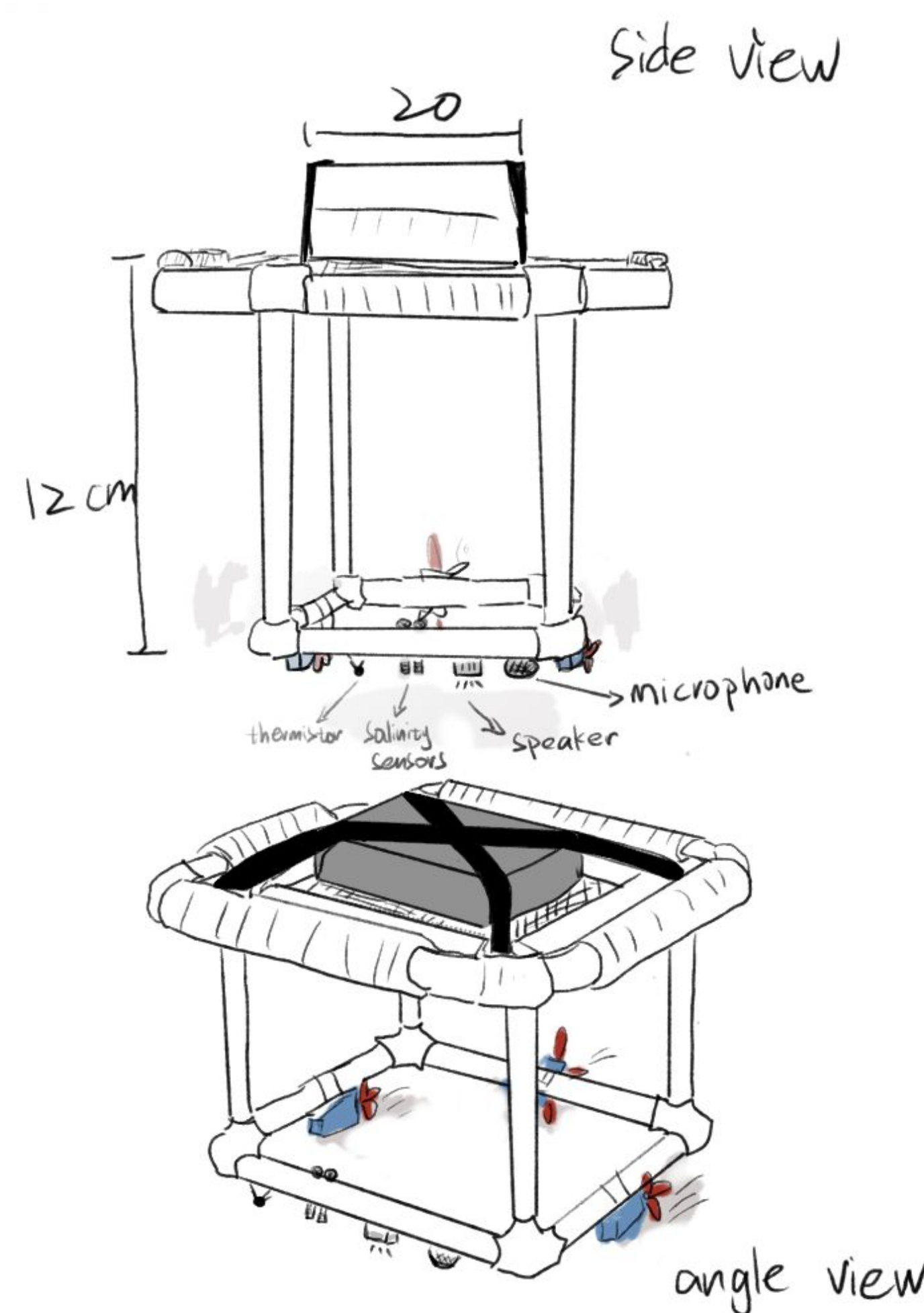
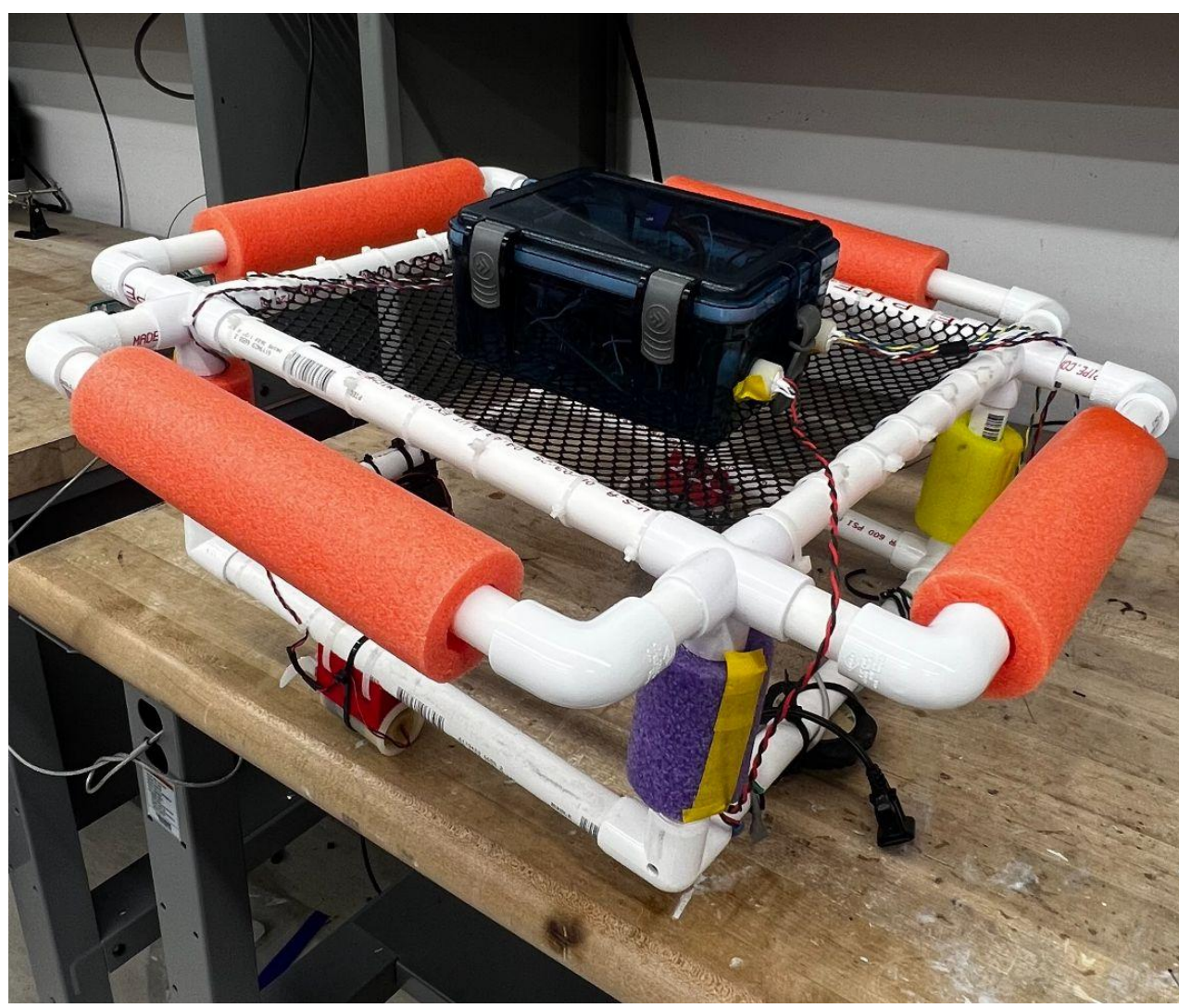
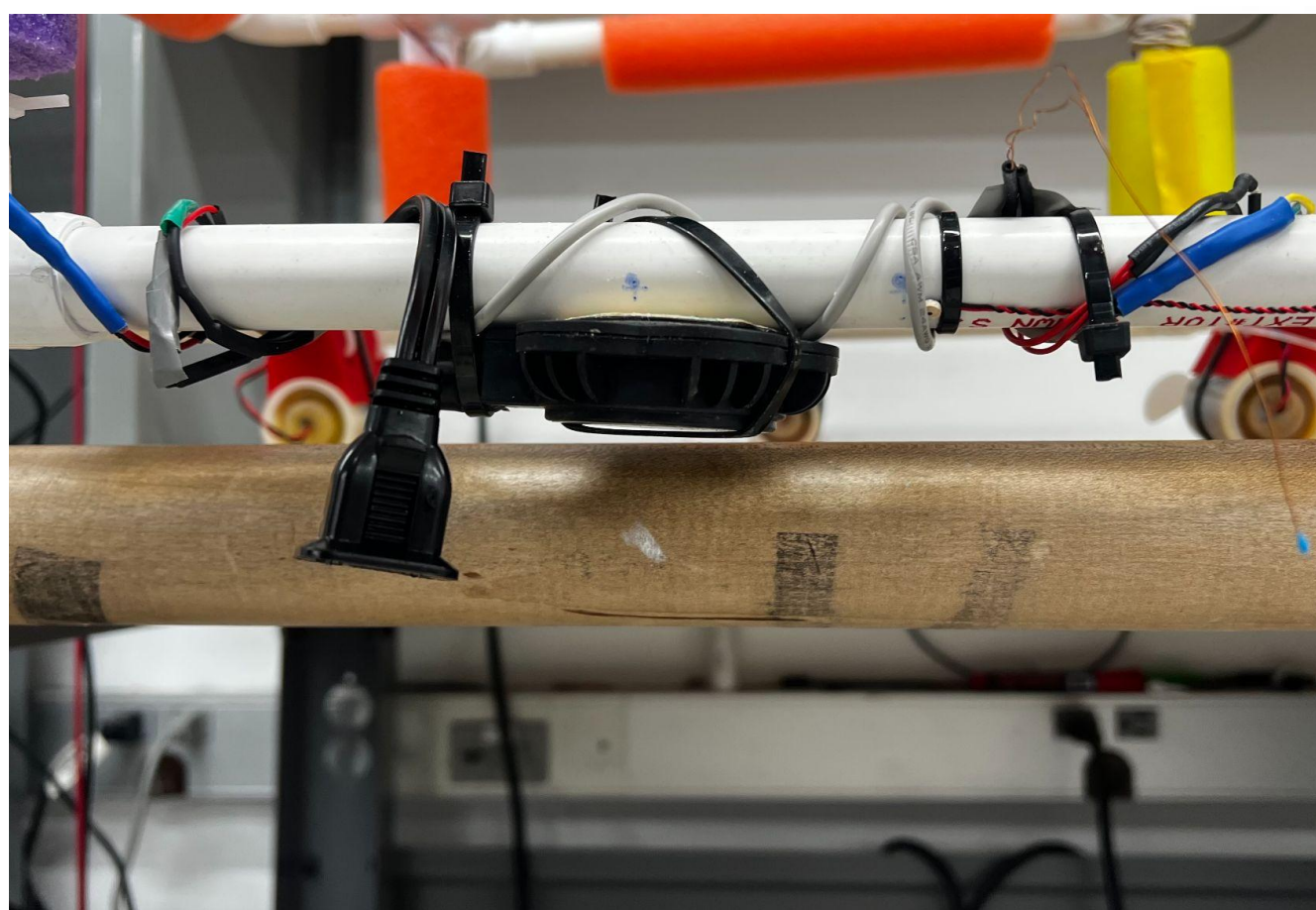


Figure 5. Mechanical design and ideation of the AUV

Experimental Data and Results

Sensor Calibration

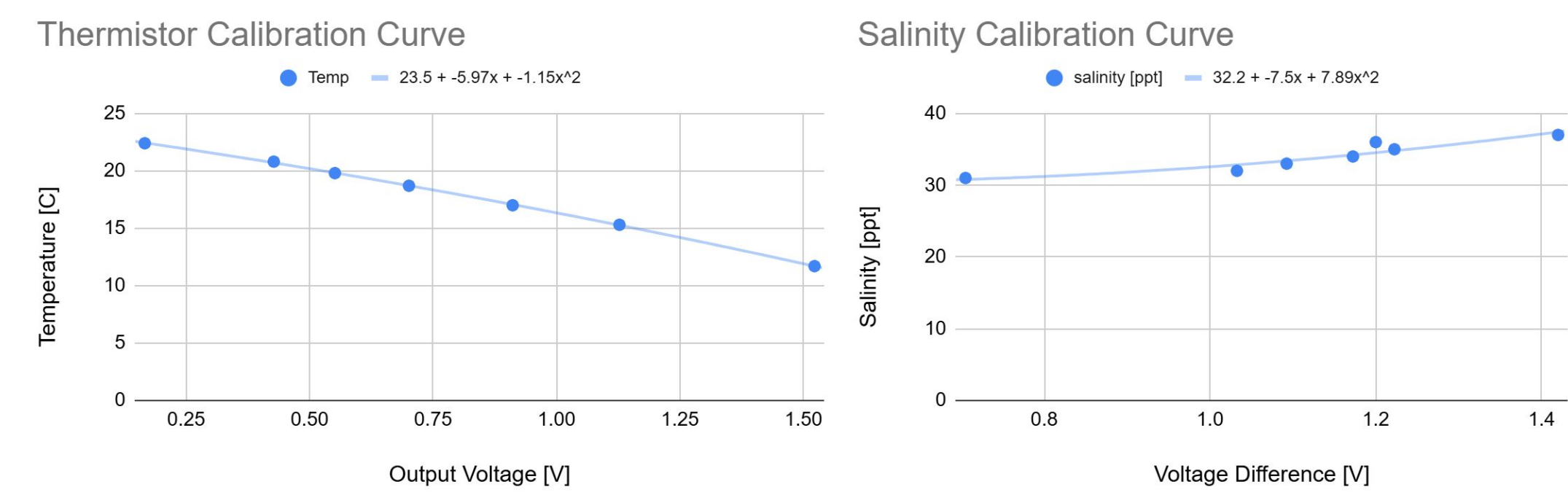


Figure 6. Temperature Calibration

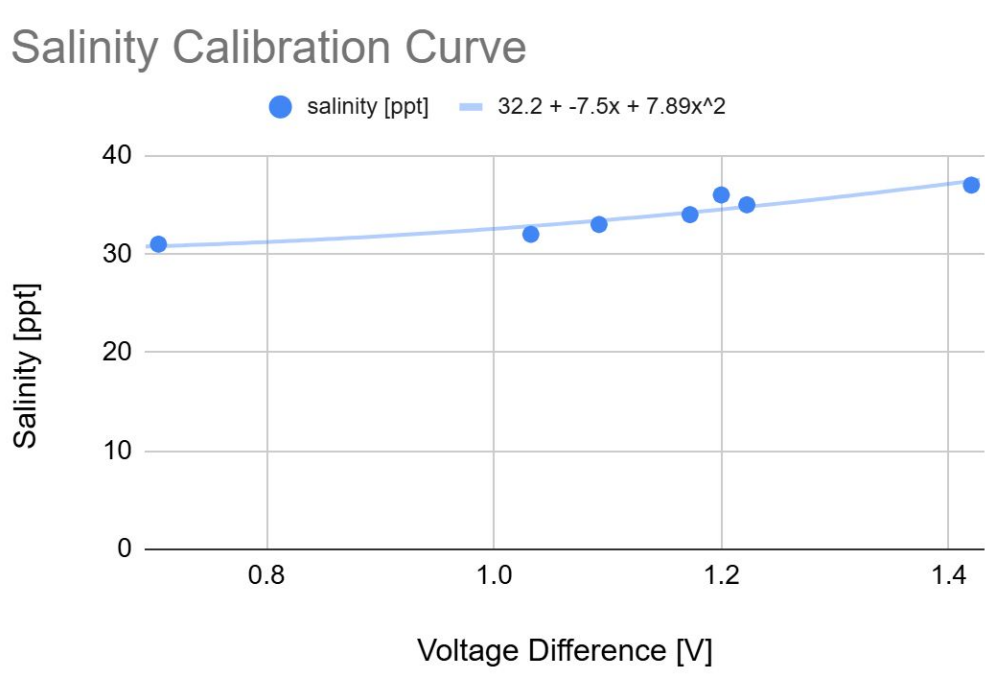


Figure 7. Salinity Calibration

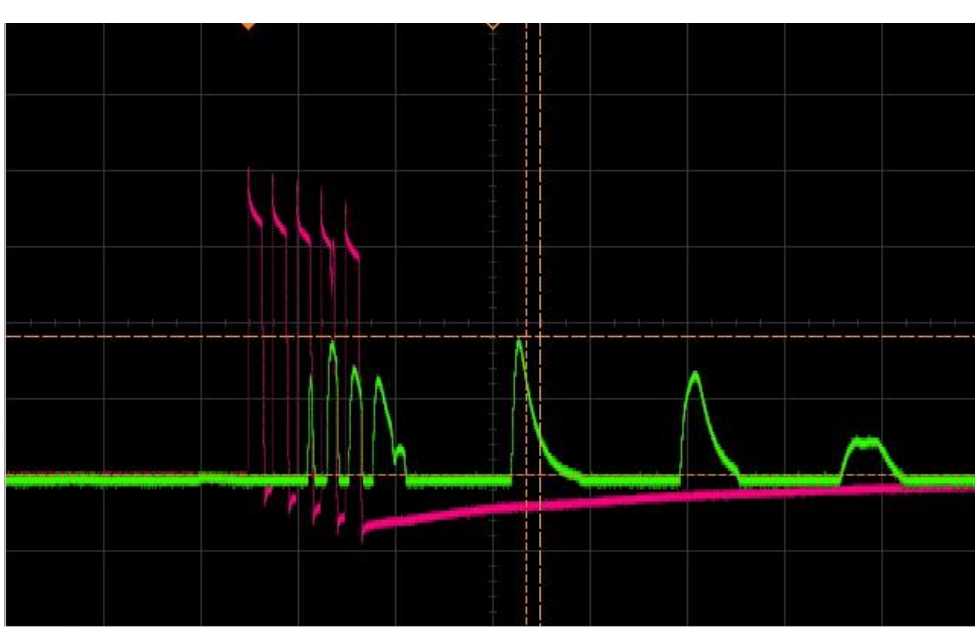


Figure 8. Sonar Validation

Ground Truth Measurements

- Salinity: Salinity data of Dana Point typically ranges from 33-35 ppt [1]
- Thermistor: Temperatures measured by a thermometer at Dana Point was 19 °C.
- Sonar: Depth measurements validated by the pHake lake, tank room, and Axelrood pool testing.

Results & Conclusions

Sonar testing by the shore at Dana Point gave us accurate depth readings of 3.93 feet. However in kayak tests we couldn't collect depth data. This is most likely due to interferences since the surface is not flat unlike our other testing in Axelrood Pool, so the microphone was unable to pick up the sound. The thermistor readings steadily decreased from 21.25 to 20.6 as the robot moved further away from the shore. The temperature at Dana Point in years past was typically around 17-18 °C, so the warmer temperature readings reflect global warming impacts. Comparing the salinity data collected from the nearshore vs. the offshore, the similarity in their salinity levels (33.5 ppt) is indicative of there not being a difference in the wastewater runoff by the shore versus our kayak testing further out in the ocean.

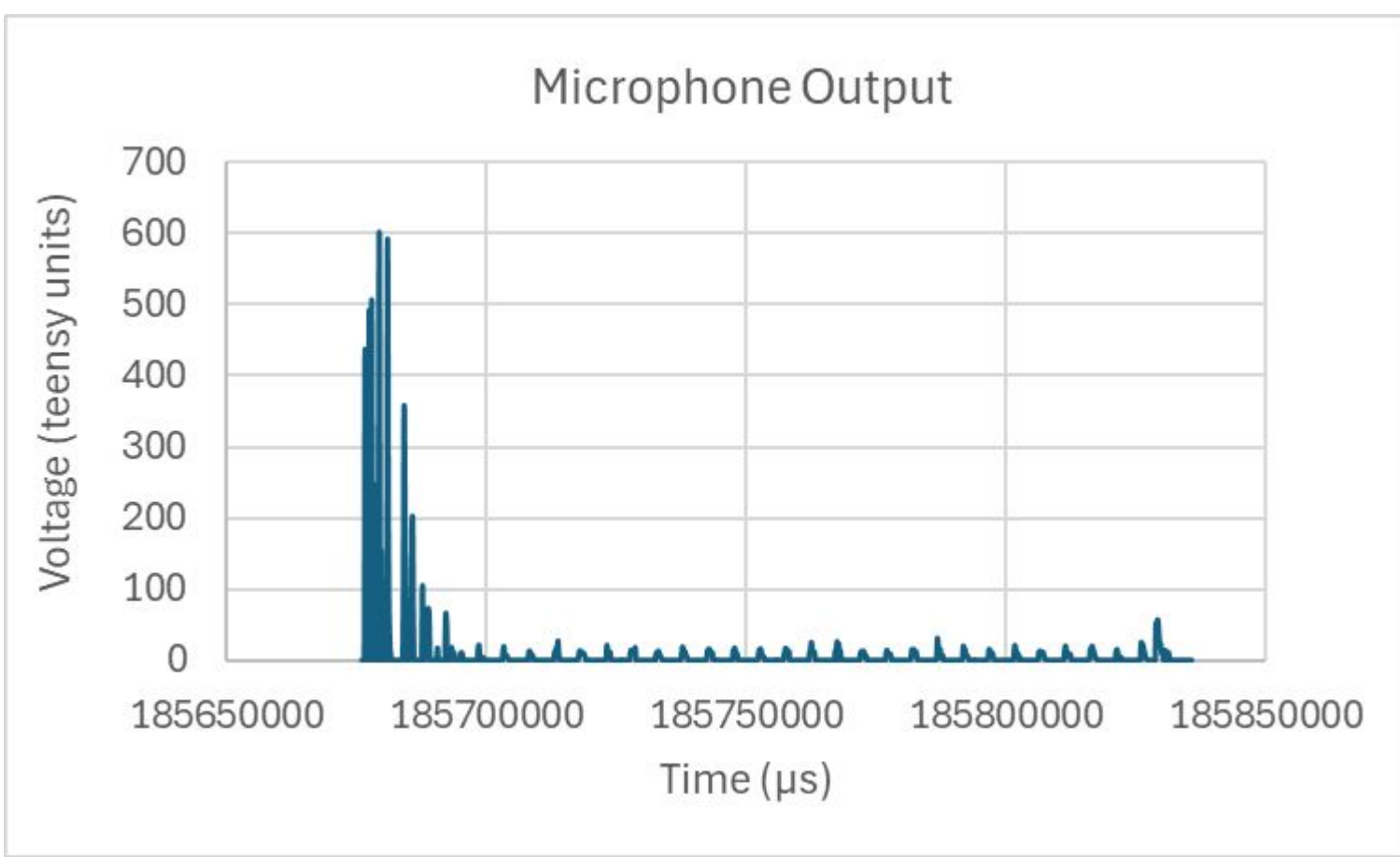


Figure 9. Plot of Microphone Output

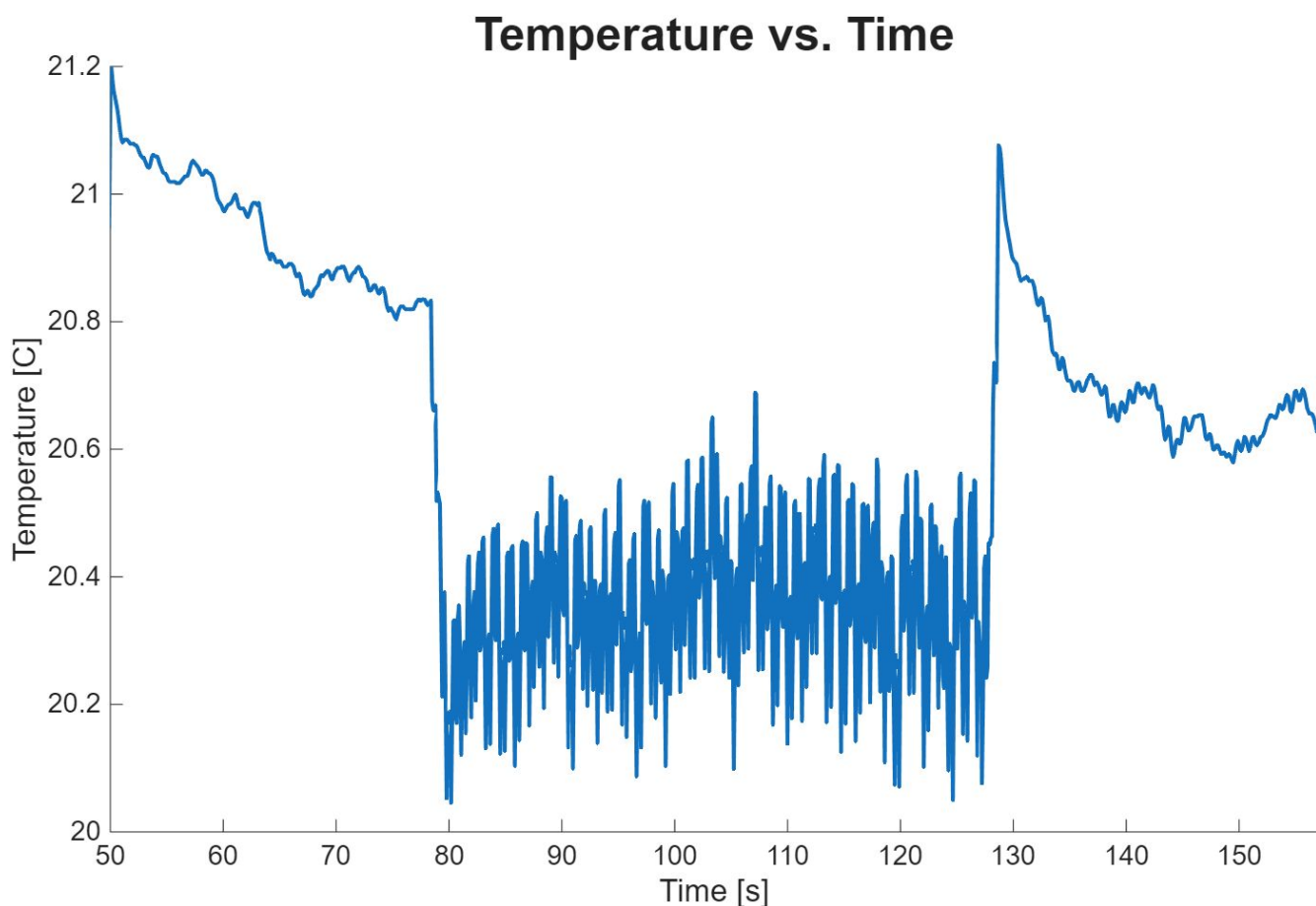


Figure 10. Graph showing Temp. vs Time

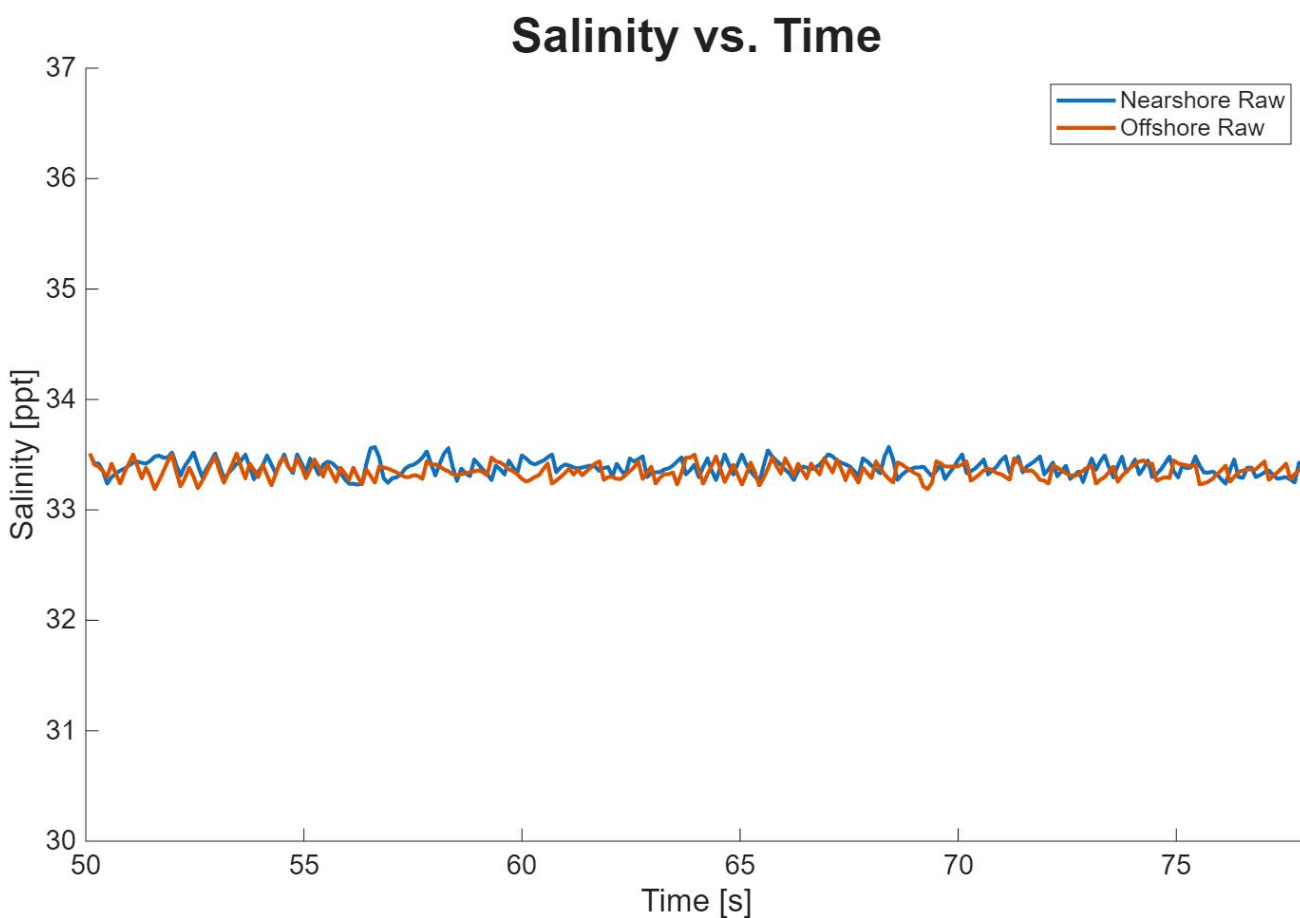


Figure 11. Graph showing Salinity vs. Time

Deployment Details

Location(s): Axelrood Pool, pHake Lake, Dana Point
Testing Protocol: Upload code, tie on tether, put on life jackets, plug in battery, seal/velcro box, place robot in water. After run is over drain water from frame, dry off box, open payload, press user switch to stop logging, remove SD card, and verify data using logging code.

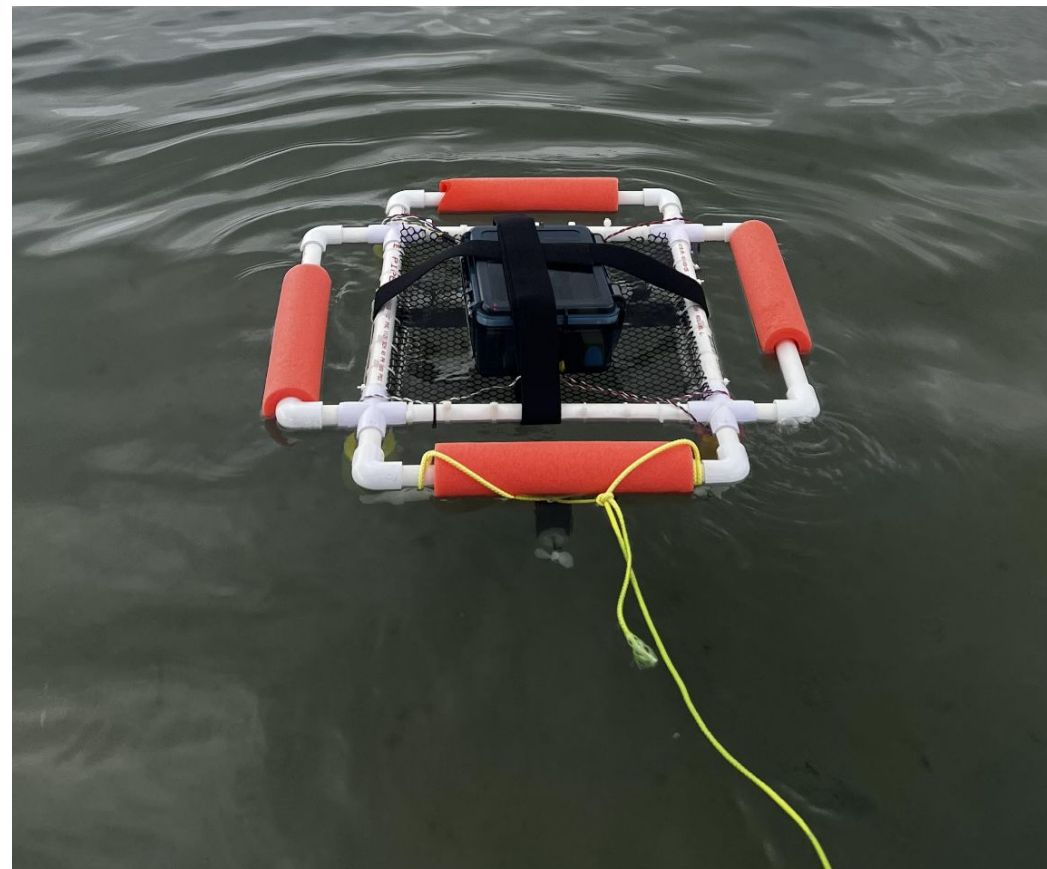


Figure 12. Picture showing robot deployment at shore

Acknowledgements

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Citations: "Dana Point Ocean Water Temperature Today | CA, United States Temp." 2020. SeaTemperature.info. 2020. <https://seatemperature.info/dana-point-water-temperature.html>. [1]